

Guide

- 1) High resolution satellite images with large regions takes too much space (in GB or TB). These large files makes it difficult to as well as slow to download, share and run.
- 2) (a) Number of tiles downloaded get quadrupled, because zoom 19 ~~is~~ would need four times as many tiles as in zoom 18. ~~to~~ to cover exact same bounding box.
(b) The resolution increases from 0.6m/pixel to 0.3m/pixel. Since each pixel captures less area, edges get sharper and model is able to find ^{small} rooftop boundaries.
- 3) If it is lowered to 0.4 then building count will increase but it will also count fake buildings.
If it is raised to ~~0.6~~ 0.9, then building count will decrease and miss ~~some~~ real buildings.
- 5) Training a model from scratch requires a lot of time as well as data to train. Since our model has learned basic things like how to detect edges, corners and general shapes, it is easier way to ~~train our~~ fine-tuning is an easier way to and faster way to train model by ~~transfer~~ weights and less data.

6) mAP compares prediction with human-made label. In dense Indian cities, human miss tiny buildings and sometimes boundaries are also not properly defined, which results in low mAP score, since it measures (IOU).

7) Since every buildings present in a image ~~are~~ ^{are} not residential, many/some of them are commercial. It will significantly affect our further predictions.

And our model is incapable to distinguish between buildings bases on their height. Hence it will also affect ~~build~~ our predictions.

8) By training a model we learn ~~and~~ ^{to use} how pre-trained heavy models work, ~~at~~ which definitely help us to rectify errors in the model and give an idea ~~at~~ how we should use them.

Using existing one is better option but ^{we} should know basic principles, pros as well as cons of that tool before using it.

Code Parib

- 1) ~~Paris~~ There is a huge difference between Paris and Indian house structure. In Paris houses are well separated from each other, have sharp edges, less dense, high quality satellite ~~img~~ images and regular building structures. On the other hand, Indian buildings are highly dense ~~with~~ and attached with each other. They do not have sharp corners and edges.
- 2) My model ~~detect~~ detected 3922 buildings. It misses where there is not sharp corners and edges, ~~highly dense~~ high building density and rooftops have irregular shapes.
- 3) Sandbox ~~to~~ model is optimized to run efficiently in a notebook environment. while ~~the~~ google trained Open Buildings on a massive scale with a huge dataset.

A ~~data~~ data scientist uses pre-trained model when there is not sufficient time or capital to make its own model. ~~And~~

It uses its own model ~~when~~ in small task with high specification.